Retriever-Conditioned Representation Search for Cross-Domain Patent-Family Retrieval

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Abstract

Citation-linked patent-family retrieval must find relevant families despite long documents, duplicate family members, and vocabulary shifts across technical domains. We test whether the candidate representation should be chosen for each frozen retriever rather than fixed for all retrievers. Retriever-Conditioned Representation Search (RCRS) evaluates a finite set of executable patent-text programs without updating retriever weights. Development results varied by retriever and did not support one universal representation. Before the single Selection-125 exposure, BM25 plus Arctic was fixed as the operational comparator. Final-872 then compared that system with the selected PatEmbed RCRS system. Recall@100 on the 872 OUT-eligible queries was 0.4425 versus 0.3311 (paired difference 0.1114; 95% bootstrap confidence interval 0.1023–0.1204; 619 wins, 158 ties, and 95 losses). This confirms a difference between two complete frozen systems, not an isolated representation effect. A later diagnosis of the unchanged winner found that 4,065 of 5,193 strictly OUT-linked relevant pairs were absent from its Top-200 pool. Under this protocol, the selected system performed better than the operational comparator, while most remaining strict-OUT error lay outside the pool available to a reranker.

Keywords: patent retrieval, patent families, representation search, cross-domain retrieval, candidate exposure, auditable evaluation

1. Introduction

A patent retrieval system rarely scores an invention as a single, uniform document. Technical content may be split across family members and across titles, abstracts, descriptions, and claims. Claims compress legal scope into specialized language; descriptions repeat concepts at several levels of detail.

Cross-domain retrieval adds vocabulary shift between the query and the relevant family. These properties make text construction part of the retrieval configuration. Field order, structural labels, passage boundaries, packing, duplicate handling, and family aggregation all affect the evidence presented to a frozen retriever.

Family-level and embedding benchmarks now support controlled study of these choices. DAPFAM defines domain-aware relevance at patent-family level [1]. PatenTEB and a broader multi-task study compare patent embeddings across retrieval, classification, and clustering tasks [2, 3]. Those studies primarily compare models or embeddings. Here the model weights stay fixed while the executable representation changes.

AutoIndex frames representation construction as a program-search problem [4]. Patent records require a more constrained grammar than generic text: the program must respect document structure, long fields, family duplication, and candidate-level aggregation. RCRS combines that deterministic patent-family grammar with retriever-conditioned staged evaluation and a frozen system-level confirmation. Optimization also needs an evaluation protocol that separates iterative development from confirmation. Repeated access to one final set would make a representation program another route for test-set adaptation.

The contribution is deliberately separated into four parts. The method contribution is an executable grammar for patent-family representations. The protocol contribution is retriever-conditioned staged selection followed by bounded confirmation. The empirical contribution is the observation that transfer outcomes varied across the tested retrievers. The diagnostic contribution is the post-confirmatory decomposition of candidate exposure from within-pool ordering. RCRS is therefore not presented as a wholly new retrieval algorithm or as a replacement for AutoIndex; it is a patent-specific representation and evaluation formulation.

Retriever-Conditioned Representation Search (RCRS) addresses both points. The method searches serializable programs that transform patent text and aggregate family evidence. Retriever weights, training data, and relevance judgments do not change. The protocol separates iterative development, one Selection-125 exposure, paired Final-872 confirmation, and post-confirmatory analysis of the unchanged winner (Fig. 1).

Repository receipts use A1–A7 for study records; retriever identifiers remain in the provenance files. The article therefore uses step names and model names in the narrative and figures; Table 5 maps the study steps back to their

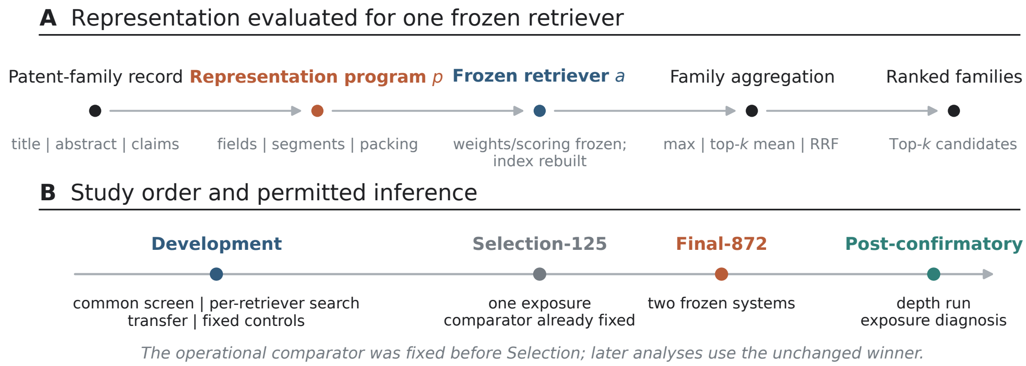

Figure 1: Method and study order. (A) An executable program constructs patent text and family-level evidence for one frozen retriever. Retriever weights and the scoring function are frozen; the derived candidate index is rebuilt deterministically for each representation program. (B) Development, Selection-125, Final-872 confirmation, and post-confirmatory analysis have separate roles. The operational comparator was fixed before Selection-125.

repository records.

The study answers three questions. First, does one representation program transfer across frozen retrievers, or are useful choices retriever-specific? Second, how does the selected frozen system compare with the prespecified BM25+Arctic operational comparator on a held-out query split? Third, after confirmation, how much strictly OUT-linked error reflects ordering inside the retrieved pool and how much reflects absent candidates? The claims remain within this family-level benchmark. We do not present a new embedding model, a component-level causal ablation, a universal baseline ranking, or a numerical comparison with external protocols that use different corpora or relevance definitions.

2. Related work

2.1. Patent-family retrieval and embedding evaluation

Patent retrieval differs from short-text web search in length, structure, duplication, and the cost of missed prior art. BM25 remains a useful lexical reference because it exposes exact-term evidence and has a well understood probabilistic basis [5]. Dense retrievers add semantic matching but impose model-specific context, pooling, and input conventions. Multilingual and patent-specialized embedding models therefore need evaluation at the same unit used by the application.

DAPFAM makes the family the evaluation unit and distinguishes cross-domain relations [1]. That design avoids counting several publications of the same invention as independent successes. PatenTEB and patent-embedding benchmarks cover a broader set of tasks and models [2, 3]. Their published scores cannot be placed in a common ranking with the present results unless corpus version, query construction, family normalization, relevance unit, and metric denominators match. They position the research question but do not serve as numerical comparators.

2.2. Executable representations, fusion, and candidate depth

An executable representation program records more than a prompt or a text template. It specifies the operations that produce indexable units and the rule that maps unit scores back to a patent family. AutoIndex provides the closest general formulation [4]. Our grammar specializes that idea to structured patent fields and ties every program to a frozen retriever.

DAPFAM evaluates patent retrieval choices such as models, document or passage representations, family aggregation, and rank fusion [1]. RCRS does not replace that benchmark or claim a new embedding or reranker. It adds a finite, serializable search over representation programs, removes duplicate executable actions, conditions each search on one frozen retriever, and separates development, one selection exposure, and confirmation. Compared with AutoIndex, the search mechanism remains program based, but the grammar is limited to patent fields, long-document units, duplicate handling, and patent-family aggregation. The main difference is therefore the patent-specific grammar and evaluation protocol, not a wholly new search algorithm.

Table 1 separates these contributions. The individual operations are not claimed as new. RCRS contributes their patent-specific, executable identity, conditions the search on a frozen retriever, and places selection and confirmation behind separate data exposures.

Optimization in a compound system can produce flat or unstable search surfaces when distinct proposals compile to the same action or when a local change is incompatible with the surrounding components [6]. The evaluation therefore records the effective program signature and negative outcomes rather than counting proposals as independent progress. Rank fusion supplies a separate test of complementarity. Reciprocal rank fusion (RRF) combines lists without calibrated scores [7], but an extra list can still add noise. Best-single, top-two, top-three, all-primary, and commercial-only controls were

Table 1: Scope of the closest frameworks. A check marks an explicit focus, not a numerical comparison across incompatible protocols.

Framework	Main unit	Executable program identity	Retriever-conditioned search	Protected selection and confirmation
DAPFAM [1]	patent-family benchmark experiments	no	evaluates multiple retrievers and representations	no
AutoIndex [4]	general representation programs	yes	frozen-retriever program optimization	not its focus
RCRS	patent-family representation programs	yes; duplicate actions removed	yes; one search per frozen retriever	yes; one selection exposure and one frozen system comparison

fixed before evaluation.

Candidate depth separates first-stage exposure from later ordering. Passage retrieval can support fine-grained patent novelty analysis [8], but a downstream reranker or novelty model cannot act on an absent family. The post-confirmatory oracle therefore asks only what perfect ordering of the already frozen Top-200 could achieve. It is an analytical bound, not a reranker result.

3. Method

3.1. Retriever-conditioned objective

Let a denote a frozen retriever, $p \in \mathcal{P}$ an executable representation program, q a query patent record, and d a candidate patent-family record. Program p emits a finite set of candidate views,

$$\mathcal{V}_p(d) = \{v_1, \ldots, v_m\}. \tag{1}$$

The frozen retriever scores each view, and the program maps those scores to a family score,

$$S_{a,p}(q, d) = G_p(s_a(q, v_1), \ldots, s_a(q, v_m)) \,. \tag{2}$$

The development search selects

$$p_a^* = \arg\max_{p\in\mathcal{P}} J_a(p), \tag{3}$$

where J_a is the prespecified family-level objective for retriever a. OUT Recall@100 is primary; latency, cost, coverage, failures, and determinism are guardrails. The subscript a is substantive: a program selected for one retriever is not assumed to transfer to another.

For max-p aggregation, the score of a family is the highest score among its emitted passages: $G_p(s_1, \ldots, s_m) = \max_i s_i$. In plain terms, a family receives the score of its best-matching passage. The query side uses the frozen benchmark query text and each retriever's locked query format. PatEmbed uses "encode query for different document retrieval:"; Arctic uses "query:"; BGE-M3 uses no instruction; and Qwen3 uses the recorded patent-family retrieval instruction. The representation program changes candidate-family views, not query membership or relevance labels.

For a query q, $R_q(q)$ denotes the query representation and $R_p(d)$ denotes the candidate-family views emitted by program p. Retriever weights and the scoring function are bound before evaluation. Changing p changes the candidate views and therefore the derived index; candidate indexes are rebuilt deterministically for each evaluated representation. The model, scoring function, relevance judgments, and metric code remain fixed.

For a query-level relevant-family set $\mathrm{Rel}(q)$, per-query recall and macro query-averaged recall are

$$r_q@k = \frac{|\,\mathrm{Rel}(q) \cap \mathrm{Top}_k(q)|}{|\,\mathrm{Rel}(q)|}, \qquad \mathrm{Recall}\,@k = \frac{1}{|Q|}\sum_{q\in Q} r_q@k. \tag{4}$$

The pair-level exposure diagnostic is different:

$$\mathrm{Exposure}\,@k = \frac{\sum_q |\,\mathrm{Rel}(q) \cap \mathrm{Top}_k(q)|}{\sum_q |\,\mathrm{Rel}(q)|}. \tag{5}$$

Macro query recall must therefore not be compared numerically with pair-level exposure.

Binary relevance is computed as

$$\mathrm{rel}(q,d) = \begin{cases} 1, & \text{if the recorded score is greater than zero,} \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

3.2. Executable grammar and program identity

The grammar contains auditable text operations. A program selects title, abstract, and claims; orders fields; adds structural labels; creates document, section, claim, passage, or multiview units; sets passage size, overlap, and boundary handling; packs units within the retriever's input limit; normalizes Unicode, whitespace, case, and punctuation; applies a duplicate policy; and aggregates evidence by maximum score, top-three mean, top-two mean, or view-level RRF. The program cannot change retriever parameters, training data, the corpus, relevance labels, or metric code.

Every candidate program is serialized. Evaluation keys use its effective action signature, so syntactically different proposals that compile to the same behavior are counted once. This rule matters for the flat search surface reported in Section 5.

3.3. Finite representation-search procedure

The search used a frozen candidate manifest rather than an unconstrained live LLM sweep. The manifest contained 52 proposals: 40 matched proposals and 12 conditional reserves. All 52 compiled to distinct executable signatures. Its proposal roles were exploit, matched-ablation, orthogonal, and diversity, and it was frozen before measurement with seed 20260812. Each frozen proposal was compiled, mapped to an effective action signature, and skipped if that signature had already been evaluated. For a new signature, the procedure constructed query and candidate views, evaluated the candidate on development data, and recorded valid, invalid, or dormant status with cost, coverage, and failures. The predeclared decision class (strict gain, numerical tie, or no strict gain) was then applied and the selected configuration was frozen. No proposal mutated a retriever, corpus, judgment, metric, or later split.

Before comparison, the primary metric is rounded to 12 decimal places with round-half-even. A strict gain means that this rounded candidate score is greater than the rounded frozen incumbent score. Equality is a numerical tie. There is no additional tolerance and no secondary-metric or latency tiebreaker. Dormant candidates remain unmeasured and do not imply a metric.

The A2 accounting was 52 frozen proposals, 44 measured proposals, 8 dormant reserves, and zero declared scientific failures.

Table 2: Finite representation-search accounting. The candidate manifest was frozen before measurement. Every proposal had a distinct executable signature; dormant reserves were not measured.

Retriever	Proposals	Unique sigs.	Measured	Dormant	Recorded outcome
BM25	8	8	8	0	diagnostic tie
BGE-M3	8	8	8	0	diagnostic tie
PatEmbed	12	12	8	4	numerical tie
Arctic	12	12	8	4	strict gain
Qwen3	12	12	8	4	no strict gain
Total	52	52	44	8	–

The separate A3 search procedure proposed 12 programs in three complete batches, but all 12 compiled to one executable action. It therefore stopped at a flat search surface. A2 and A3 are separate development records.

3.4. Frozen retrievers

All retrievers were frozen before their reported evaluations. They span a lexical baseline, a general multilingual embedding model, a patent-specialized model, and two recent general embedding models (Table 3). Context limits are recorded configuration properties; they do not imply equal effective coverage of a patent record. PatEmbed uses CC BY-NC-SA 4.0, which may constrain commercial deployment. The license is therefore part of the system boundary, not a footnote to the effectiveness result.

The frozen BM25S configuration used $k_1 = 1.2$, $b = 0.75$, Lucene-style scoring, and a NumPy backend. Its tokenizer applied Unicode NFKC normalization and case folding, then extracted word tokens; it used neither stemming nor stopword removal.

Both systems used the same fixed Final-872 query scope and candidate-family pool, but their retrievers, query encoders or formats, scoring functions, input limits, and indexes were system specific and frozen before confirmation. Final-872 therefore compares these two complete systems. It does not isolate representation and is not a claim against all static representations.

4. Experimental design

4.1. Benchmark and evidence stages

Every experiment belongs to the same family-level cross-domain benchmark lineage [1]. The benchmark construction is publication/source records $\rightarrow$ benchmark family records $\rightarrow$ ordered text fields $\rightarrow$ normalization and

Table 3: Frozen retrievers. Short names are used in the text and figures; repository retriever identifiers remain in the provenance files.

Short name	Retriever	Type; dimension; context	License	Role in development
BM25	BM25S 0.3.10	lexical; –; –	–	diagnostic lexical baseline [5]
BGE-M3	BAAI/bge-m3	dense multilingual; 1,024; 8,192	MIT	diagnostic [9]
PatEmbed	datalyes/ patembed-large	patent dense; 1,024; 512	CC BY-NC-SA 4.0	advanced; research/non-commercial
Arctic	Snowflake Arctic-Embed-M-v2.0	dense multilingual; 768; 8,192	Apache 2.0	advanced [10]
Qwen3	Qwen3-Embedding-0.6B	dense multilingual; 1,024; 32,768 model-declared tokens	Apache 2.0	advanced [11]

Table 4: The two systems compared on Final-872. They use different retriever compositions and candidate representations. The result is therefore a comparison of two complete frozen systems, not an isolated representation effect. Verification identifiers are listed in Appendix Table A.8.

System	Human-readable binding
Selected RCRS	PatEmbed only; title, abstract, claims; 384-token passages; 64-token overlap; NFKC whitespace normalization; content-hash-first duplicate policy; max-p family aggregation
Operational comparator	BM25 plus Arctic, synchronous, depth 100; title, abstract, claims; one family document; NFKC canonical whitespace with case preserved; all source members retained; single-unit family aggregation

duplicate policy → views/passages → one family score. The complete retrieval pool contains 45,336 candidate families. Table 5 records population, exposure, freeze state, and permitted interpretation for each study step. Development is iterative. Selection-125 is the only selection exposure. Final-872 compares exactly two frozen systems. The depth run and candidate-exposure diagnosis use only the confirmed winner after Final-872.

Table 5: Study steps, populations, exposure, and permitted interpretation. Repository codes identify experiment records, not retrievers.

Record	Study step	N	OUT population	System state and exposure	Permitted inference
A1–A3	Development: common screen, per-retriever search, transfer, and controls	250	development-specific	five frozen retrievers; iterative search; fixed transfer and fusion controls	describe retriever-specific outcomes, transfer, ties, and negative controls
A4	Selection-125	125	OUT-eligible $n = 90$	four frozen profiles; one exposure	select the research reference
A5	Final-872 confirmation	872	eligible cohort; 17,429 judged positive pairs	two frozen systems; no tuning; paired by query	compare RCRS and the operational comparator
A6	Full-benchmark depth run	1,247	relation-scoped	confirmed winner only; depth 200; after Final-872	report a depth stress test only
A7	Candidate-exposure diagnosis	905	strict IN/OUT; 5,193 positive OUT pairs	unchanged winner and frozen Top-200 pool	separate exposure from within-pool ordering

4.2. Patent-family construction and split protection

Each query is an English patent-family record represented by its title, abstract, and claims. The system ranks 45,336 candidate families, and positive relations are those with relevance score greater than zero. The benchmark labels each positive relation as IN or OUT by its domain relation. The recorded aggregate evidence does not expose a separate temporal-control rule, so no temporal-independence claim is added here.

The benchmark treats a candidate family as the unit of retrieval and evaluation. Publication/source records are grouped into benchmark-defined family records; the recorded comparator program retains all source members and orders them by publication ordinal and opaque publication token. It renders the title, abstract, and claims fields with NFKC canonical whitespace and case preservation, then forms one family document. The selected RCRS program uses title, abstract, and claims fields with NFKC whitespace normalization, 384-token passages, 64-token overlap, content-hash-first duplicate handling, and max-p family aggregation. The frozen specifications do not separately declare a language-selection filter or a near-duplicate rule; neither is inferred here. These details distinguish the two frozen specifications without exposing member identifiers or protected split payload.

The split unit is the benchmark query record. Development comprises the REP-DEV 150-query cohort, the HDEV-100 100-query cohort, and their combined Train-250 250-query development record. Selection-125 contains 125 queries, with OUT metrics on the verified (n=90) eligible subset. Final-872 contains 872 paired queries. Split membership and judgments were fixed before exposure and remain protected. A local aggregate-only audit found zero shared query records and zero duplicate normalized title–abstract–claims text across Development, Selection-125, and Final-872. The available metadata does not map query records to query-family or source-publication members, so the study establishes query-split separation but does not claim family-level independence.

4.3. Development, transfer, and fixed controls

The common screen applies one representation to all five retrievers. The per-retriever search then preserves each recorded decision class (strict improvement, tie, or no strict improvement). The transfer study applies each advanced-retriever program to every advanced target retriever without editing. The 3×3 matrix tests compatibility; it is not a second selection procedure.

Five fixed controls test whether rank-list complementarity changes the result: the strongest single system, top-two RRF, top-three RRF, all-primary fusion, and a commercial-only profile. The all-primary and top-three controls resolve to the same aggregate scores in the frozen record. None of these development controls can replace Selection-125 or enter Final-872 after inspection.

4.4. Selection and confirmation

Selection-125 evaluates four frozen profiles at depth 100. The research reference is PatEmbed alone. The operational comparator combines BM25 and Arctic synchronously. BALANCED combines BM25, Arctic, and Qwen3 synchronously; DEEP uses the same three retrievers asynchronously. The primary OUT subset contains 90 queries. The research reference won all three pairwise decisions on eligible-cohort Recall@100 after every profile passed full coverage, zero-failure, and determinism checks. The exact comparator system was registered as the static/common baseline before Selection-125 was opened. The registry had zero selection accesses and bound its system hash, synchronous mode, depth 100, and commercial-capable license scope. Selection identified the research champion; it did not select or modify the comparator. The resulting pair was then frozen for Final-872. Confirmation requires 872/872 coverage and paired query-level comparison; a failure would remain visible in the report.

4.5. Metrics, uncertainty, and population accounting

Recall@k is computed at patent-family level. Recall@100 on the OUT-eligible query cohort is primary. nDCG@100 is a confirmatory secondary outcome. nDCG@10 is descriptive because the canonical Final-872 artifact contains no confidence interval for that endpoint. nDCG uses the benchmark's binary relevance grades. Final-872 differences are paired by query. Percentile 95% confidence intervals use 10,000 paired bootstrap resamples of the paired query units; win, tie, and loss counts use the same query unit. These intervals describe uncertainty from resampling the 872 evaluated queries under this fixed protocol; they are not uncertainty intervals for all patents or future search settings. Development decision classes are strict gain, numerical tie, and no strict gain, as recorded in the aggregate ledger.

The notation distinguishes two denominators. In Selection-125, an OUT-eligible query has at least one positive relation whose `domain_rel` value is OUT. All 125 queries are ranked, but only the 90 eligible queries enter OUT metric vectors and paired statistics; the other 35 are not inserted as zeroes. Final-872 selects queries by OUT eligibility but evaluates all judged positive families for those queries, not only relations whose `domain_rel` is OUT. Its metric denominator contains 17,429 positive query–family pairs. OUT_{strict} is used only in the post-confirmatory full-benchmark diagnosis: raw self-relations are removed, then metrics are computed on the 905 queries that retain at least one positive OUT relation, comprising 5,193 positive pairs. IN and OUT can overlap at query level because one query may have both relation types. These populations serve different analyses and are not a before–after series.

5. Development results

5.1. Common screen and per-retriever search

Common-screen Recall@100 was 0.1912 for BM25, 0.2699 for BGE-M3, 0.4134 for PatEmbed, 0.3407 for Arctic, and 0.3637 for Qwen3. The corresponding nDCG@100 values were 0.1727, 0.2314, 0.3478, 0.2845, and 0.3079. BM25 and BGE-M3 remained diagnostic; PatEmbed, Arctic, and Qwen3 entered advanced development. Appendix Table A.7 gives the full common-screen and per-retriever aggregate ledger.

The common-screen and per-retriever values are separate development records; the first is not the incumbent used for the second decision. Against their frozen A2 incumbents, Arctic scored 0.3587 versus 0.3527 and met its strict-gain rule, PatEmbed tied at 0.4230, and Qwen3 scored 0.3737 versus 0.3800 and did not meet the rule. BM25 and BGE-M3 remained diagnostic. Program outcomes therefore varied across the tested retrievers without implying an A1-to-A2 gain from the side-by-side display.

5.2. Transfer, fusion, and effective search surface

Figure 2A reports all nine advanced-retriever transfers. The three programs produced a narrow Recall@100 range on PatEmbed (0.4184–0.4193), a lower range on Arctic (0.3374–0.3413), and 0.3595–0.3626 on Qwen3. No program source held a consistent diagonal advantage, and the target-retriever ordering remained visible after transfer.

The diagonal cells in Fig. 2A are descriptive transfer cells, not the per-retriever winner values in Appendix Table A.7. The transfer matrix and the ledger use different development cohort and configuration summaries: the matrix applies each source program unchanged to a target retriever, whereas the ledger reports the separate per-retriever search record. Their diagonal values therefore need not agree, and no value is corrected or promoted from one record to the other.

The controls were also non-monotonic (Fig. 2B). Best-single and top-two RRF were nearly tied in Recall@100 (0.4184 and 0.4187), while top-two RRF increased nDCG@100 from 0.3471 to 0.3527. All-primary fusion and top-three RRF both returned 0.4151 Recall@100 and 0.2848 nDCG@10, below the strongest single system on those endpoints. The commercial-only profile returned 0.3693 Recall@100. It quantifies a licensing trade-off within this benchmark; it is not an alternative Final winner.

The separate search procedure proposed 12 candidates in three batches, but their effective action signatures collapsed to one. Additional proposal rounds did not create additional executable behavior. Nominal proposal count would therefore overstate the amount of search; the effective signature is the reported unit.

6. Selection and confirmatory results

6.1. One selection exposure

All four Selection-125 profiles completed 125/125 queries. The research reference obtained 0.4161 eligible-cohort Recall@100. BALANCED and DEEP tied at 0.3606; FAST obtained 0.3083. Against either BALANCED or DEEP, the paired difference was 0.0556 (95% CI 0.0300–0.0817; 49 wins, 20 ties, 21 losses). Against FAST, the difference was 0.1078 (95% CI 0.0811–0.1350; 69/9/12). The recorded Selection-125 result contains no verified nDCG value, so none is inferred. The reported intervals and win/tie/loss counts are descriptive selection diagnostics, not confirmatory inference, because this exposure was used for profile selection. FAST is reported here as the registered profile name. It was the prespecified static/common operational comparator, not the strongest commercial-capable system observed at Selection-125. Selection fixed the RCRS champion and did not alter the comparator; the final pair was then evaluated without tuning.

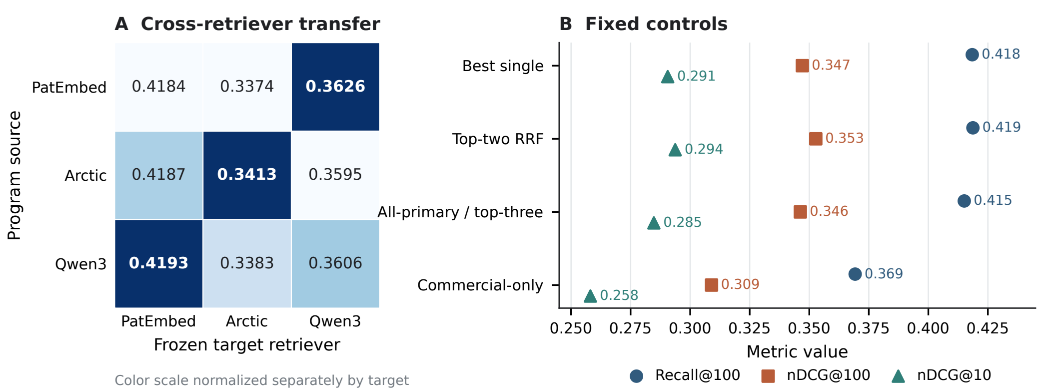

Figure 2: Development-only transfer and fixed controls. (A) Recall@100 for programs optimized on PatEmbed, Arctic, or Qwen3 and applied to each frozen target retriever; shading is normalized within a target column, so the darkest cell marks only that column's highest transfer score. (B) Fixed fusion and licensing controls. These data were not used to revise the Final-872 systems.

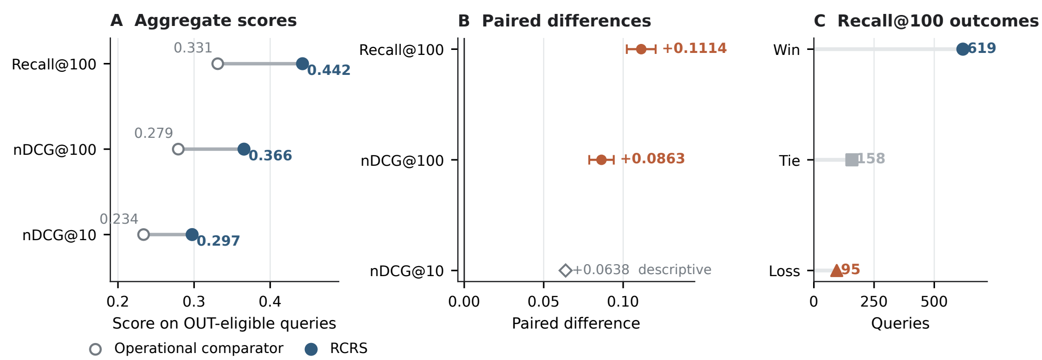

Figure 3: Final-872 comparison of two frozen systems. (A) Aggregate effectiveness on the OUT-eligible query cohort; all judged positive families for each eligible query enter the metric. (B) Paired query-level differences and 95% bootstrap confidence intervals; nDCG@10 is descriptive. (C) Query-level Recall@100 wins, ties, and losses.

6.2. Final-872 effectiveness

Both Final-872 systems completed 872/872 queries without failure. RCRS obtained 0.4425 Recall@100 on the OUT-eligible query cohort; the operational comparator obtained 0.3311 (Fig. 3). The paired difference was 0.1114 (95% CI 0.1023–0.1204), with 619 wins, 158 ties, and 95 losses. nDCG@100 increased from 0.2793 to 0.3656, a paired difference of 0.0863 (95% CI 0.0787–0.0941). Descriptive nDCG@10 increased from 0.2337 to 0.2975, an aggregate difference of 0.0638.

Under the frozen Final-872 protocol, the selected PatEmbed RCRS system outperformed the BM25+Arctic operational comparator. The estimate is a difference between two complete frozen systems. It does not show that representation alone caused the difference. External systems and individual grammar components were outside the design.

The operational results were mixed. RCRS reduced median latency and had a similar p95, but its p99 was 1,044.613 ms versus 804.002 ms for the operational comparator. Throughput was higher for RCRS (3.319344 versus 2.665918 queries per second), a direct ratio of 1.245. These are recorded observations from the frozen run, not an independently reproducible efficiency benchmark: the available aggregate record does not define cache state, repetition count, or the exact timing boundary. They are not evidence of patent-office deployment scalability, and the worse tail latency prevents an unqualified efficiency claim.

Table 6: Final-872 effectiveness on OUT-eligible queries and recorded operational checks. Metrics include all judged positive families for each eligible query. Confidence intervals are paired query-level bootstrap intervals. The p99 latency result is retained although it favors the comparator.

Outcome	Operational comparator	RCRS	Difference or ratio	Statistical or operational note
Eligible-cohort Recall@100	0.3311	0.4425	+0.1114	CI 0.1023–0.1204; 619/158/95
Eligible-cohort nDCG@100	0.2793	0.3656	+0.0863	95% CI 0.0787–0.0941
Eligible-cohort nDCG@10	0.2337	0.2975	+0.0638	descriptive; no verified interval
Latency p50 / p95 (ms)	315.862 / 735.171	237.547 / 732.218	–	lower median; similar p95
Latency p99 (ms)	804.002	1,044.613	–	higher for RCRS
Throughput	2.665918 q/s	3.319344 q/s	1.245	direct throughput ratio
Coverage / failures	872/872; 0	872/872; 0	–	both deterministic in the recorded run

7. Post-confirmatory diagnosis

7.1. Complete-benchmark depth profile

After Final-872 fixed the selected RCRS configuration, the full-benchmark depth run evaluated that system alone on 1,247 queries and 45,336 candidate families to depth 200. The run produced 249,400 ranked rows with no failure. Overall Recall@10, @20, @50, @100, and @200 was 0.1368, 0.2147, 0.3364, 0.4390, and 0.5468 (Fig. 4A). IN recall at those depths was 0.1870, 0.2779, 0.4131, 0.5282, and 0.6451. Strict OUT recall was 0.0340, 0.0709, 0.1340, 0.1884, and 0.2602. nDCG@100 was 0.3625 overall, 0.4065 for IN, and 0.0706 for OUT_{strict}.

The depth run contains no comparator and uses relation-scoped denominators. It describes the frozen selected RCRS configuration at increasing candidate depth; it does not re-estimate the Final-872 effect.

7.2. Strict OUT exposure and Top-200 bound

The candidate-exposure diagnosis partitions 5,193 OUT_{strict}-relevant family pairs (Fig. 4B). Of these, 796 appeared in ranks 1–100, 332 only in ranks 101–200, and 4,065 were absent at rank 200. Among 905 judged OUT_{strict} queries, 67 were fully retrieved by rank 200, 297 were partially retrieved, 86 had relevant evidence only in ranks 101–200, and 455 had no relevant candidate in the Top-200 pool. Appendix Table A.9 records these counts.

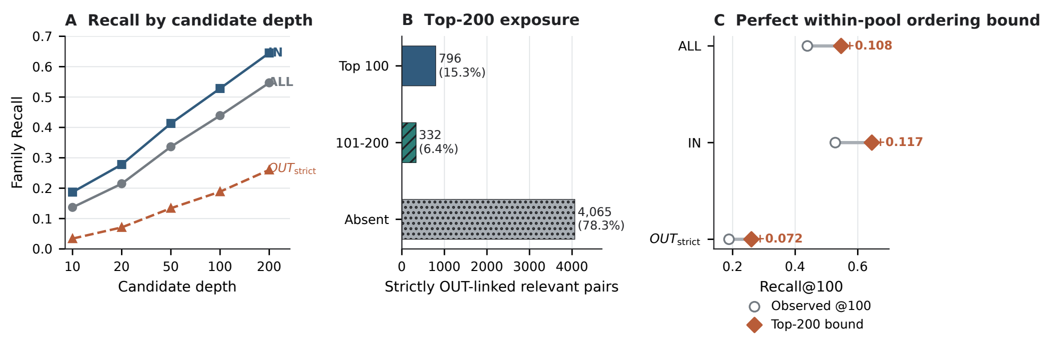

Figure 4: Diagnosis of the unchanged selected RCRS configuration after confirmation. (A) Macro query-averaged family recall by candidate depth. (B) Pair-weighted exposure counts for 5,193 OUT_{strict}-relevant pairs. (C) Macro query-averaged Recall@100 and the analytical bound from perfect ordering of the frozen Top-200 pool. IN and OUT are relation-scoped and may overlap by query.

The analytical oracle moves each relevant family already present in the frozen Top-200 into the first 100 positions; it adds no candidate. The resulting Recall@100 bounds were 0.5468 overall, 0.6451 for IN, and 0.2602 for OUT_{strict}, compared with observed values of 0.4390, 0.5282, and 0.1884. The within-pool gaps were 0.1079, 0.1169, and 0.0717 (Fig. 4C). These are ordering bounds, not achieved reranker scores. Even perfect ordering of the exposed Top-200 would leave 4,065 strict OUT_{strict}-relevant pairs unavailable.

The macro recall curves give each query equal weight. The exposure counts give each relevant query–family pair equal weight, so a query with many relevant families contributes more pairs. The two summaries answer different questions and need not have the same numerical value.

8. Discussion

8.1. Representation is part of the retriever configuration

The development results varied across retrievers. Arctic met its strict-improvement rule; PatEmbed and Qwen3 did not. Transfer changed scores by target retriever and produced no consistent source-program advantage. These outcomes fit a conditional view: the useful text units and aggregation rule depend on the scoring behavior and input constraints of the retriever that consumes them.

Final-872 shows that the selected frozen RCRS system can outperform the prespecified BM25+Arctic operational comparator. The estimate is paired, its interval excludes zero, both systems cover the same 872 queries, and the comparator was fixed before the selection exposure. The negative and tied development outcomes show that improvement was not observed for every retriever, proposal batch, or fusion.

8.2. Candidate exposure is a separate engineering target

The exposure decomposition narrows the next question. A reranker can reorder the 332 strictly OUT-linked relevant pairs found only in ranks 101–200 and can change the placement of relevant pairs already in the first 100. It cannot recover the 4,065 pairs absent from the Top-200. Candidate generation, coverage, and cross-domain exposure therefore need separate measurement from ordering quality.

The oracle is reported as a bound. Calling the Top-200 bound a reranking result would imply an implemented model and an attainable gain. Neither exists here. The bound measures only the largest Recall@100 value permitted by the frozen candidate pool.

8.3. Implications for patent-search evaluation

Three reporting practices follow from the evidence. A representation program should be versioned with its retriever, not listed as generic preprocessing. Family normalization, duplicate handling, passage packing, and score aggregation should appear in the executable configuration. The benchmark's citation-derived relevance relations are evaluation labels, not legal findings about novelty or inventive step. Finally, relation-scoped populations should be reported with their query and pair denominators. The same label, such as OUT, is insufficient when protocols use different eligibility rules.

For search practice, the results separate four decisions. Family grouping and duplicate handling prevent repeated publications of one invention from being counted as separate retrieval successes. Candidate depth sets a hard ceiling: a later reranker cannot repair a false negative that never enters its pool. Representation must therefore be tested with the retriever that will consume it, and licensing must be considered before a research system is treated as a deployment option.

Licensing also constrains system choice. PatEmbed supports the research comparison but its non-commercial share-alike terms may exclude a production deployment. The commercial-only control quantifies one in-benchmark alternative, but it does not establish the best deployable system. A production study would need a new prespecified comparison among models with compatible terms.

8.4. Limitations and follow-up work

The evidence is bounded in six ways. It comes from one family-level benchmark with 45,336 candidate families, not a patent-office-scale deployment. The benchmark-scale latency and throughput measurements should not be read as production capacity evidence. Citation-derived relevance is not a legal novelty or inventive-step determination. Final-872 contains one frozen multi-retriever comparator, not an exhaustive baseline suite or a same-retriever representation ablation. The grammar changes several components jointly, and no hash-bound component ablation identifies a causal field, segmentation rule, or aggregation operation. External benchmark configurations were not verified as comparable, so cross-study scores are not ranked. IN and OUT are relation-scoped and can overlap by query. The candidate-exposure diagnosis contains no implemented rescue model or layer-three reranker.

The aggregate audit found no shared query records or normalized query text, but the available metadata cannot test query-family or source-member overlap across development, Selection-125, and Final-872. The public package also omits restricted split data, complete execution assets, and per-query results. These limits weaken independent verification and should be resolved before a strong held-out-family or public-reproducibility claim is made.

A separate preregistered study could test grammar ablations, additional licensed baselines, an independent benchmark, and candidate-generation methods with a new development/final split. Reusing Final-872 for those decisions would weaken the confirmation reported here.

9. Conclusion

On Final-872, the selected frozen PatEmbed RCRS system exceeded the prespecified BM25+Arctic operational comparator by 0.1114 in Recall@100 on the OUT-eligible query cohort. This difference is specific to the two tested systems and the family-level protocol. After confirmation, the unchanged selected RCRS configuration exposed only 1,128 of 5,193 OUT_{strict}-relevant pairs in its Top-200, placing most of the remaining error outside the pool available to a reranker. The system comparison and the exposure limit are separate: the tested RCRS system performed better than the operational comparator, while broader candidate coverage remains an open retrieval problem.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Data and code availability

The benchmark remains subject to its source terms, and PatEmbed remains subject to CC BY-NC-SA 4.0. The submission package contains aggregate tables, machine-readable aggregate figure inputs, deterministic figure-generation code, serialized representation specifications, frozen configuration identifiers, an aggregate-only split-overlap audit, and provenance paths for every reported number. It is internally auditable and configuration-bound, but it is not independently reproducible from the public artifacts alone. It does not redistribute query identifiers, patent or family identifiers, relevance judgments, per-query outcomes, or raw rankings. The blinded repository URL, archive identifier, and final availability statement will be added after owner review.

Declaration of generative AI and AI-assisted technologies

During manuscript preparation, the authors used OpenAI Codex for language editing, document structure, LaTeX implementation, and quality-control support. The authors reviewed and edited all output, checked every reported result against the study records, and take full responsibility for the article.

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Appendix A. Aggregate evidence ledger

The appendix is part of the main manuscript. No supplementary scientific file is required. Tables A.7 and A.9 contain aggregate values only.

Table A.7: Key common-screen (repository A1) and per-retriever search (repository A2) aggregates. Decision classes compare the A2 candidate with its frozen A2 incumbent, not with the common-screen score. Scores are rounded to 12 decimal places before strict comparison.

	Common-screen record		A2 record		
Retriever	R@100	nDCG@100	Candidate R@100	Incumbent R@100	Decision class
BM25	0.191200	0.172717	0.234667	–	diagnostic
BGE-M3	0.269933	0.231377	0.290000	–	diagnostic
PatEmbed	0.413400	0.347812	0.423000	0.423000	numerical tie
Arctic	0.340667	0.284546	0.358667	0.352667	strict gain
Qwen3	0.363733	0.307930	0.373667	0.380000	no strict gain

Table A.8: Verification identifiers for the two Final-872 systems. These identifiers support local checking but are not part of the scientific comparison.

System	SHA-256 prefixes
Selected RCRS	winner binding `285e94b1...`; representation `bcbffffa...`; retrieval `e2ba107f...`
Operational comparator	system `9a98a6d8...`; program `00f1c295...`; query format `51a81f1b...`

Table A.9: Candidate-exposure diagnosis (repository A7). The strict OUT pair-level total is 5,193; the judged-query total is 905.

Pair-level class	Count	Query-level class	Count
Relevant pair in Top-100	796	fully retrieved by rank 200	67
Relevant pair in ranks 101–200	332	partially retrieved by rank 200	297
Relevant pair absent at rank 200	4,065	deep-only at ranks 101–200	86
		no relevant candidate in Top-200	455